

Case A

two forward paths

three single loops

no non-touching loops

$$\Delta = 1 - (L_1 + L_2 + L_3)$$

$$T = \frac{P_1\Delta_1 + P_2\Delta_2}{\Delta}$$

Case B

two forward paths

three single loops

two pairs of non-touching loops

$$\Delta = 1 - (L_1 + L_2 + L_3) +$$

(pair products)

add positive product terms